

M KIRTHIK ADITYA

Cybersecurity & Cloud Security Engineer

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PROFESSIONAL SUMMARY

Cybersecurity and Cloud Security enthusiast with hands-on experience across Linux, Docker, AWS, networking, and penetration testing. Built cloud-based cybersecurity lab infrastructure and a secure data-sanitization tool using cryptographic erasure and HSM-backed key management. Comfortable across the stack in Go, Python, and Bash. Focused on Cloud Security, DevSecOps, and Infrastructure Security.

PROJECTS

BIT CyberSpace — Cybersecurity & Cloud PaaS Platform

Technologies: Go, Docker, Linux, Nginx, Wireshark, Nmap, Cloud Infrastructure

- Developed backend services in Go to power a cloud-based cybersecurity lab platform, enabling students to run penetration tests against containerized environments and analyze PCAP files to capture flags.
- Hosted 12+ containerized lab environments supporting 20 concurrent users, with purpose-built vulnerable environments for hands-on exploitation practice.
- Built and managed Docker-based containerized environments on Linux, following secure configuration practices to isolate lab workloads from the host and from each other.
- Configured Nginx for reverse proxy, TLS termination, and service routing, cutting external attack surface by 40% and centralizing access control to backend containers.
- Integrated Wireshark- and Nmap-based workflows for traffic capture/analysis and reconnaissance exercises, and designed the architecture around cloud security principles (network segmentation, least-privilege access), reducing environment provisioning time by 80% through deployment automation.

ZeroTrace — Secure Data Erasure & Media Sanitization Tool

Technologies: Python, Bash, HSM, Cryptographic Erasure, Bootable ISO

- Built a bootable ISO and companion application for securely wiping HDDs and SSDs across 3 storage form factors, rendering data non-recoverable to support secure decommissioning of on-premise and cloud storage devices.
- Implemented cryptographic erasure as the core sanitization method — destroying the encryption key rather than overwriting data — significantly cutting wipe time versus traditional multi-pass overwrite methods (e.g., DoD 5220.22-M).
- Integrated a Hardware Security Module (HSM) for secure key generation, storage, and destruction, strengthening the trust boundary of the erasure process and preventing key recovery after sanitization.
- Developed the wiping engine and automation tooling in Python and Bash, validated across 20 test drives with a 100% successful-erasure rate, for compliance-driven asset retirement in cloud and enterprise environments.

TECHNICAL SKILLS

Programming	Go, Python, Bash Scripting
Cloud	AWS, IAM & Access Control
Containers & DevOps	Docker, Linux, Git/GitHub
Security	Wireshark, Burp Suite, Nmap, Metasploit, Vulnerability Assessment, Penetration Testing
Networking	TCP/IP, DNS, HTTP, OSPF, Packet Analysis

EDUCATION

B.Tech – Artificial Intelligence & Machine Learning

Bannari Amman Institute of Technology | 2024 – 2028

CERTIFICATIONS & AREAS OF INTEREST

- Cisco – Introduction to Cybersecurity
- Pwn Till Dawn — online.pwntilldawn.com/Achievements/9259
- CyLab Academy — learn.cylabacademy.org/users/spryzz3n

Interests: Cloud Security | DevSecOps | Network Security | Cybersecurity Engineering

CTF & SECURITY COMMUNITY ACTIVITY

- Active on tryhackme.com/p/SPRYZZEN — solving rooms and CTF-style challenges across web, network, and cloud security tracks.
- Publish cybersecurity and cloud security write-ups on medium.com/@mkirthikaditya.al24, translating hands-on lab and project work into technical blog posts.
- Hands-on practice in Linux administration, networking, and cloud environments through personal homelabs, with code and configs on github.com/spryzz3n.